Chap 5.

Q2.

(i) $\sigma = (1\,4\,3)(2\,3)(2\,4)$,

Note these three are <u>not</u> disjoint.

So to compute σ will mean to find its expression as a product of disjoint cycles.

Two cycles are disjoint when the symbols mentioned in them are disjoint

$$\sigma = (3\,2)(1\,4)$$

$\sigma(3)$ $\sigma(2)$

(ii) $\sigma = (1\,2\,3)(4\,5)(1\,2\,5\,4)^{-2}$.

$$= (1\,4\,3)(2\,5)$$

$$\sigma = (1\,2\,3)(4\,5)\left((1\,2\,5\,4)^{-1}\right)^{2}$$

$$= (1\,2\,3)(4\,5)(1\,4\,5\,2)^{2}$$

$$= (1\,2\,3)(4\,5)(1\,4\,5\,2)(1\,4\,5\,2)$$

(k) ~~ie~~

~~If~~ For $g \in G$

$|g| =$ least positive $k > 0$ such that $g^k = e$.

$|(1\ 2\ 5\ 4)| = 4$

(j) $(1\ 2\ 5\ 4)^{100} = \left((1\ 2\ 5\ 4)^4\right)^{25}$

$= ((1))^{25}$

$= (1)$, the identity.

$(1\ 2\ 5\ 4)^{113} = (1\ 2\ 5\ 4)^{112+1}$

$= \left((1\ 2\ 5\ 4)^4\right)^{28} (1\ 2\ 5\ 4)$

$= (1\ 2\ 5\ 4)$

Mock 02.

Q1.

(a) For $a, b \in \mathbb{Z}$.

we say a divides b, $a \mid b$, if and only if $\exists\, q \in \mathbb{Z}$. $b = aq$.

Let $d = \gcd(a,b)$ and consider S.

$$S = \{ xa + yb : x, y \in \mathbb{Z}, xa + yb > 0 \}.$$

~~Let~~ By the "well-ordered" principle of $\mathbb{Z}$.

S has a least element $\delta \in S$.

Claim δ is a common divisor of a, b.

Proof Proof by contradiction.

Suppose $\delta \nmid a$. Consider the integer division

$$a = q\delta + r, \quad 0 < r < \delta$$

$$\Rightarrow r = a - q\delta$$

$$= a - q(xa + yb), \text{ for some } x, y \in \mathbb{Z}$$

$$= (1 - qx)a - qyb \text{ as } \delta \in S.$$

$\Rightarrow \quad r \in S$

This contradicts the fact that δ is the least element of S.

Therefore $\delta \mid a$.

Similarly $\delta \mid b$.

So δ is a common divisor of a, b and $\delta = xa + yb$, for some $x, y \in \mathbb{Z}$.

and so if c is any other common divisor of a, b then $c \mid \delta$ so $c \leq \delta$

So $\delta = \gcd(a, b)$, i.e. $\delta = d$.

$= xa + yb$, as required.

(b). Using proof by induction

Base case when $n = 1$ says that

$13 \mid 5^4 - 1$

$\Leftrightarrow \quad 13 \mid 625 - 1$

$5^3 = 125$

$5^4 = 625$

$\Leftrightarrow 13 \mid 624$ 624

Which is true as $624 = 48 \cdot 13$

So assume that $13 \mid 5^{4k} - 1$, for some $k \geq 1$. Consider the next statement.

$$5^{4(k+1)} - 1$$

$$= 5^{4k+4} - 1$$

$$= 5^4 \cdot 5^{4k} - 1$$

$$= 5^4 (5^{4k} - 1) + (5^4 - 1).$$

This is a linear combination of integers both divisible by 13 (base case and induction assumption) and therefore is divisible by 13

$$\Rightarrow 13 \mid 5^{4(k+1)} - 1$$

and so $\forall n \geq 1$ $13 \mid 5^{4n} - 1$, by induction.

Assume $n > 0$

$A \Rightarrow B \equiv \neg B \Rightarrow \neg A$

d). Claim

$2^n - 1$ is prime $\Rightarrow$ n is prime.

<u>Proof</u>. We will prove the contra-positive.

claim n is composite $\Rightarrow$ $2^n - 1$ is composite.

Assume n is composite.

$\Rightarrow$ $n = a \cdot b$, where $a, b \in \mathbb{Z}^+$, $a, b \neq 1$

$a, b > 0$

$$2^n - 1 = 2^{ab} - 1 = (2^a)^b - 1^b$$

$$= \underbrace{(2^a - 1)}_{\text{since } a>1,\ 2^a - 1 > 1.}\underbrace{(2^{a(b-1)} + 2^{a(b-2)} + \dots + 2^a + 1)}_{\text{since } b>1,\ (\ \ ) > 1.}.$$

$$\Rightarrow 2^n - 1 = \underline{\qquad\qquad} \times \underline{\qquad\qquad}$$

Therefore $2^n - 1$ is composite.

see GIMPS website.

(Continue at 5:05)

Q2.

$$(a^2 - b^2) = (a-b)(a+b)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$\vdots$$

$$a^n - b^n = (a-b)\left(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \dots + ab^{n-2} + b^{n-1}\right)$$

Q2 $\mathbb{Z}_2 = \{0, 1\}$

$+$	0	1
0	0	1
1	1	0

$\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0,0), (0,1), (1,0), (1,1)\}$

(i) For groups $(G, *)$, $(H, \circ)$ the operation in the external direct product $G \times H$ is

$\forall (g_1, h_1), (g_2, h_2) \in G \times H.$

$$(g_1, h_1)(g_2, h_2) := (g_1 * g_2, h_1 \circ h_2)$$

$\mathbb{Z}_2 \times \mathbb{Z}_2$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,0)$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,1)$	$(0,1)$	$(0,0)$	$(1,1)$	$(1,0)$
$(1,0)$	$(1,0)$	$(1,1)$	$(0,0)$	$(0,1)$
$(1,1)$	$(1,1)$	$(1,0)$	$(0,1)$	$(0,0)$

$\mathbb{Z}_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

(iv) Comparing these tables we see in $\mathbb{Z}_2 \times \mathbb{Z}_2$ every element is its own inverse, but in $\mathbb{Z}_4$

1 and 3 are inverses of each other.
So $\mathbb{Z}_2 \times \mathbb{Z}_2$ and $\mathbb{Z}_4$ are <u>not</u> isomorphic.

(c). $GL(n, \mathbb{Q})$ consist of all $n \times n$ matrices with rational entries that are invertible, i.e. with non-zero determinant.
~~The~~ To prove $GL(n, \mathbb{Q})$ is a group under matrix multiplication we note.

1. Closure.
By the definition of matrix multiplication if $A, B \in GL(n, \mathbb{Q})$ then all entries in AB are again rational

and since $\det(AB) = \det(A)\det(B)$
$\det(AB) \neq 0$, as $\det(A) \neq 0$ & $\det(B) \neq 0$.

So $AB \in GL(n, \mathbb{Q})$.

2. Associativity. From linear algebra matrix multiplication is known to be associative.

(can prove it from the definition

$$(AB)_{i,j} = \sum_{k=1}^{n} A_{ik} B_{kj} \quad)$$

3. The identity element is the identity matrix I (in the lin. alg sense)

$$I = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}.$$

4. The inverse elements will be the "inverse matrices" (in the lin alg. sense).

Note if $A \in GL(n, \mathbb{Q})$

Then $\det(A) \in \mathbb{Q}$ as $\det(A)$ is a big expression of sums and products of entries of A. and so A^{-1} exists as $\det(A) \neq 0$ and moreover $A^{-1} \in GL(n, \mathbb{Q})$ as the cofactor definition of A^{-1} preserves

rationality of entries.

d). $H = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : \begin{array}{l} a, b, c \in \mathbb{Q} \\ ac = \det\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \neq 0 \end{array} \right\}$.

We'll use the subgroup theorem to prove H is a subgroup of $GL(2, \mathbb{R})$

Let $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}, \begin{pmatrix} d & e \\ 0 & f \end{pmatrix} \in H$.

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}\begin{pmatrix} d & e \\ 0 & f \end{pmatrix} = \begin{pmatrix} ad & ae+bf \\ 0 & cf \end{pmatrix}.$$

Note all four entries are in $\mathbb{Q}$ as $a, b, c, d, e, f \in \mathbb{Q}$. and $adcf \neq 0$

since $ac, df \neq 0$.

So the product is in H, ie. H is closed under the op.

The identity element $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in H$.

as it meets the def. $a = c = 1$, $b = 0$,

For inverses, note

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}^{-1} = \frac{1}{ac}\begin{pmatrix} c & -b \\ 0 & a \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{a} & -b/ac \\ 0 & \frac{1}{c} \end{pmatrix} \in H$$

since all entries are in $\mathbb{Q}$, as $a, b, c \in \mathbb{Q}$. and $\frac{1}{ac} \neq 0$ as $ac \neq 0$.

So by the subgroup theorem, H is a subgroup of $GL(2, \mathbb{Q})$.

Q3.

(b) Claim: $a^{-1} \pmod{n}$ exists $\iff \gcd(a, n) = 1$.

Prove both directions seperately.

Assume a^{-1} exists mod n.

so $a \cdot a^{-1} \equiv 1 \pmod{n}$

$\implies n \mid a \cdot a^{-1} - 1$, by def. of $\equiv$

$\Rightarrow a \cdot a^{-1} - 1 = q^n$, for some $q \in \mathbb{Z}$, by def. of divisibility

$\Rightarrow 1 = a \cdot a^{-1} - q^n$

so any common divisor of a, n must divide 1. Therefore $\gcd(a, n) = 1$.

Conversely.

Assume $\gcd(a, n) = 1$

$\Rightarrow \exists\, x, y \quad ay + n \cdot x = 1$, from the Extended Euclidean Algorithm.

$\Rightarrow ay - 1 = n \cdot x$

$\Rightarrow n \mid ay - 1$, def of divisibility

$\Rightarrow ay \equiv 1 \pmod{n}$, def of $\equiv$

So there is an inverse for a mod n namely $a^{-1} \equiv y \pmod{n}$.

To find 13^{-1} mod 100 we run the extended Euclidean Algorithm

on $(13, 100)$

$$100 = 7 \cdot 13 + 9$$
$$13 = 1 \cdot 9 + 4$$
$$9 = 2 \cdot 4 + 1$$

so $\gcd(13, 100) = 1$ as expected.

$$1 = 9 - 2 \cdot 4$$
$$= 9 - 2 \cdot (13 - 9)$$
$$= 3 \cdot 9 - 2 \cdot 13$$
$$= 3 \cdot (100 - 7 \cdot 13) - 2 \cdot 13$$
$$= 3 \cdot 100 - 23 \cdot 13$$

$$\Longrightarrow \quad 13^{-1} \equiv -23 \pmod{100}$$
$$\equiv 77$$

So $y = 77$.